Welcome to Data 498D! Capstone for Data Science

Joe Watkins & Lianfen Qian

Department of Mathematics

08/24/2026

THE UNIVERSITY OF ARIZONA

Goals for Today

- Get to know your classmates and instructional team
- Learn more about the course and the expectations you should have for the semester
- Have several engaging discussions

Introduce Yourself!

- Who are you?
  - Name
  - Pronouns
- What are you studying?
  - Major
  - Field in which you want to apply your major
- When are you graduating?
- Where do you want to end up?
  - In 5-7 years, what will you be doing?
  - Why are you passionate about it?
- How are you going to achieve your goals?

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Class Overview – Your Capstone Experience

- Transform from students who can run analyses into **professionals who deliver real impact with data**
  - Master Professional Tools
  - Become a confident communicator
  - Build your professional calling card

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Class Overview – How This Class Works: A "Flipped" Approach

- **Before Class (Your Prep Time):**
  - Complete assigned readings
  - Come ready to discuss, ask questions, and dive deeper
- **During Class (Active Learning Time):**
  - Discuss readings and concepts together
  - Work through hands-on activities and exercises
  - Collaborate with classmates on problems
  - Get immediate feedback and clarification
- **Outside Class (Project Time):**
  - Apply what you've learned to substantial projects
  - Work at your own pace on larger assignments
  - Develop your portfolio pieces

What This Means for You:

- **Limited passive lectures** - you're actively engaged the whole time
- You get **real-time support** when working through challenging concepts
- More **flexible project work** outside the classroom

You'll spend class time actually *using* statistical and data science skills rather than just hearing about them. This mirrors how you'll work in your career - reading, researching, and learning independently, then collaborating and problem-solving with colleagues.

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Class Overview – Grades

- Short Assignments (30% of overall grade)
  - Class discussions
  - In class activities
  - Career center activities
- Project 1 (20% of overall grade)
  - Individual
  - Grade divided into two parts
    - Written Reflection on previous project (50% of project grade)
    - Poster (50% of project grade)
- Project 2 (30% of overall grade)
  - Small groups (2 or 3 students)
  - Grade divided into 4 parts
    - Proposal (5% of project grade)
    - Midterm Report (15% of project grade)
    - Final Report (30% of project grade)
    - Slides and Presentation (50% of project grade)
- Portfolio (20% of overall grade)
  - Submitted at end of semester

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Concepts we will be discussing this semester

Professional Development

- Handshake
- LinkedIn
- Cover Letters
- Resume Writing
- Applications
- Written Communication
- Verbal Communication
- Team Collaboration
- Portfolios

Technology Tools

- LaTeX
- Github
- Linux
- Cloud Computing
- High Performance Computing
- Python and R
- Markdown

Critical Concepts

- Ethics
- Reproducible Research

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What other topics you would like to get more familiar with this semester?

Discussion – What makes data professional?

You've all done statistical analyses before. How many of you have had to explain your results to someone who *wasn't* a statistics student?

- What was challenging about that?
- What do you wish you had known how to do better?

In your chosen field [industry/government/academia], who will you need to convince with data?

- What will those audiences care about most?
- How is that different from turning in a homework assignment?

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Reminders and Announcements

Wednesday’s Class

- Topic: Portfolios
- Homework:
  - Complete
    - PreTechnology Survey if you haven’t already
    - Posttechnology Survey
  - Read
    - How to Create a Project Portfolio for Data Science Job Applications
  - Review
    - Data Scientist’s Portfolios from the Career Foundry Blog Post
      - Identify common themes and ideas that you would want to include in your portfolio
  - Bring
    - Any digital items you want to include in your portfolio

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For Wednesday

- Take the post technology survey (We know you did one! This one is different.)
- Read *How to Create a Project Portfolio for Data Science Job Applications*
- Be prepared to brainstorm what you would like in your portfolio
  - What projects have you created?
  - What was the final product? presentation, report, program (markdown file)
  - What do you want in your portfolio?
  - Pick one project and discuss how you would improve it.